

LEADER, SK, Canada

WMO: 714590

Lat: 50.910N

Lon: 109.503W

Elev: 676

StdP: 93.47

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.6	-28.8	-36.4	0.1	-32.5	-32.4	0.2	-28.8	13.4	-1.9	12.2	-4.0	2.8	250	0.720

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	32.4	17.1	30.4	16.7	28.2	16.0	19.3	27.7	18.2	26.3	17.3	25.2	4.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	12.8	21.9	15.4	11.9	20.8	14.3	11.1	19.9	57.9	27.7	54.2	26.3	51.1	25.4	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.1	10.5	9.3	-36.1	35.9	2.9	1.9	-38.2	37.2	-39.9	38.3	-41.5	39.4	-43.6	40.7		
			-36.3	21.4	2.9	1.2	-38.3	22.3	-40.0	23.0	-41.6	23.7	-43.7	24.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.1	-11.8	-10.1	-3.1	5.0	11.3	15.8	19.0	18.3	13.0	5.1	-3.4	-10.3
	DBStd	12.70	9.00	8.10	8.27	5.24	4.55	3.46	3.09	3.51	4.64	5.28	7.33	8.46
	HDD10.0	3069	677	564	406	161	40	2	0	0	22	164	403	629
	HDD18.3	5302	936	798	664	399	222	91	28	44	169	410	653	888
	CDD10.0	931	0	0	1	11	79	176	281	258	111	13	1	0
	CDD18.3	121	0	0	0	0	3	14	51	44	8	0	0	0
	CDH23.3	2184	0	0	1	10	126	258	776	761	243	9	0	0
	CDH26.7	791	0	0	0	1	28	70	292	312	88	1	0	0
Wind	WSAvg	4.5	4.3	4.2	4.7	5.1	5.1	4.6	4.1	4.0	4.4	4.6	4.4	4.3
Precipitation	PrecAvg	365	18	17	19	28	42	59	52	43	36	16	15	20
	PrecMax	484	41	52	48	94	94	163	113	112	96	59	40	40
	PrecMin	270	2	3	1	3	5	16	12	4	0	0	3	3
	PrecStd	52	11	12	11	20	20	37	29	27	28	15	9	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	7.3	17.2	24.8	29.7	31.6	34.9	35.1	32.7	24.7	16.3	9.3
		MCWB	4.0	3.9	8.8	11.5	14.3	16.8	18.6	17.2	15.8	12.7	8.4	4.3
	2%	DB	4.5	4.3	13.1	21.0	26.7	28.6	32.2	32.6	29.2	20.2	11.8	5.7
		MCWB	1.9	1.8	6.4	9.9	13.2	15.8	18.3	16.5	14.8	10.8	6.1	2.6
	5%	DB	2.8	2.7	10.0	17.9	24.1	26.3	29.9	30.4	26.0	17.0	8.6	3.6
		MCWB	1.0	0.9	5.0	8.4	12.2	15.3	17.9	16.4	13.6	9.2	4.2	1.2
	10%	DB	0.7	0.9	6.9	15.0	21.3	24.1	27.6	27.8	22.9	13.9	5.7	1.2
		MCWB	-0.8	-0.6	3.3	7.0	11.0	14.4	17.0	15.8	12.4	7.6	2.5	-0.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	4.1	9.4	12.5	15.7	19.5	21.5	19.7	16.8	13.4	8.8	5.1
		MCDB	7.2	6.7	16.1	22.1	24.6	27.0	30.0	28.5	28.4	22.8	15.4	8.5
	2%	WB	2.2	2.1	7.1	10.6	14.1	17.6	20.0	18.5	15.3	11.3	6.5	2.8
		MCDB	4.2	3.8	12.3	19.5	24.5	25.0	28.4	27.5	26.8	18.8	11.3	5.4
	5%	WB	1.0	0.9	5.2	9.0	12.9	16.4	18.9	17.5	14.1	9.5	4.5	1.2
		MCDB	2.7	2.6	9.3	16.8	22.3	23.8	27.1	26.3	24.0	15.9	8.1	3.6
	10%	WB	-0.8	-0.5	3.4	7.5	11.7	15.2	17.9	16.6	12.9	8.0	2.7	-0.5
		MCDB	0.7	0.8	6.8	14.1	20.0	22.0	25.5	25.3	21.8	13.5	5.3	1.2
Mean Daily Temperature Range	5% DB	MDBR	10.5	10.7	11.0	13.6	14.2	12.8	14.5	15.1	14.6	13.1	10.7	10.2
		MCDBR	11.6	11.1	15.2	19.5	19.1	17.0	18.7	20.4	21.0	18.3	14.5	11.7
		MCWBR	9.7	9.0	9.3	10.3	8.6	7.2	7.4	7.6	9.1	10.0	9.4	9.0
	5% WB	MCDBR	11.8	10.9	14.3	17.8	16.5	14.5	15.9	16.5	18.6	17.1	13.6	11.6
		MCWBR	10.1	8.9	9.3	9.8	7.7	6.8	7.3	6.9	8.4	9.8	9.2	9.0
Clear-Sky Solar Irradiance	taub		0.271	0.262	0.292	0.351	0.376	0.369	0.376	0.393	0.338	0.298	0.267	0.246
	taud		2.321	2.437	2.452	2.297	2.271	2.355	2.357	2.265	2.406	2.495	2.470	2.412
	Ebn at Noon		762	893	920	888	878	887	873	835	853	834	778	755
	Edh at Noon		64	74	89	118	127	118	116	121	93	70	55	50
All-Sky Solar Radiation	RadAvg		1.19	2.21	3.56	4.84	5.83	6.12	6.62	5.35	3.88	2.37	1.26	0.89
	RadStd		0.13	0.20	0.27	0.37	0.39	0.45	0.35	0.42	0.34	0.28	0.12	0.10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	-0.38	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page